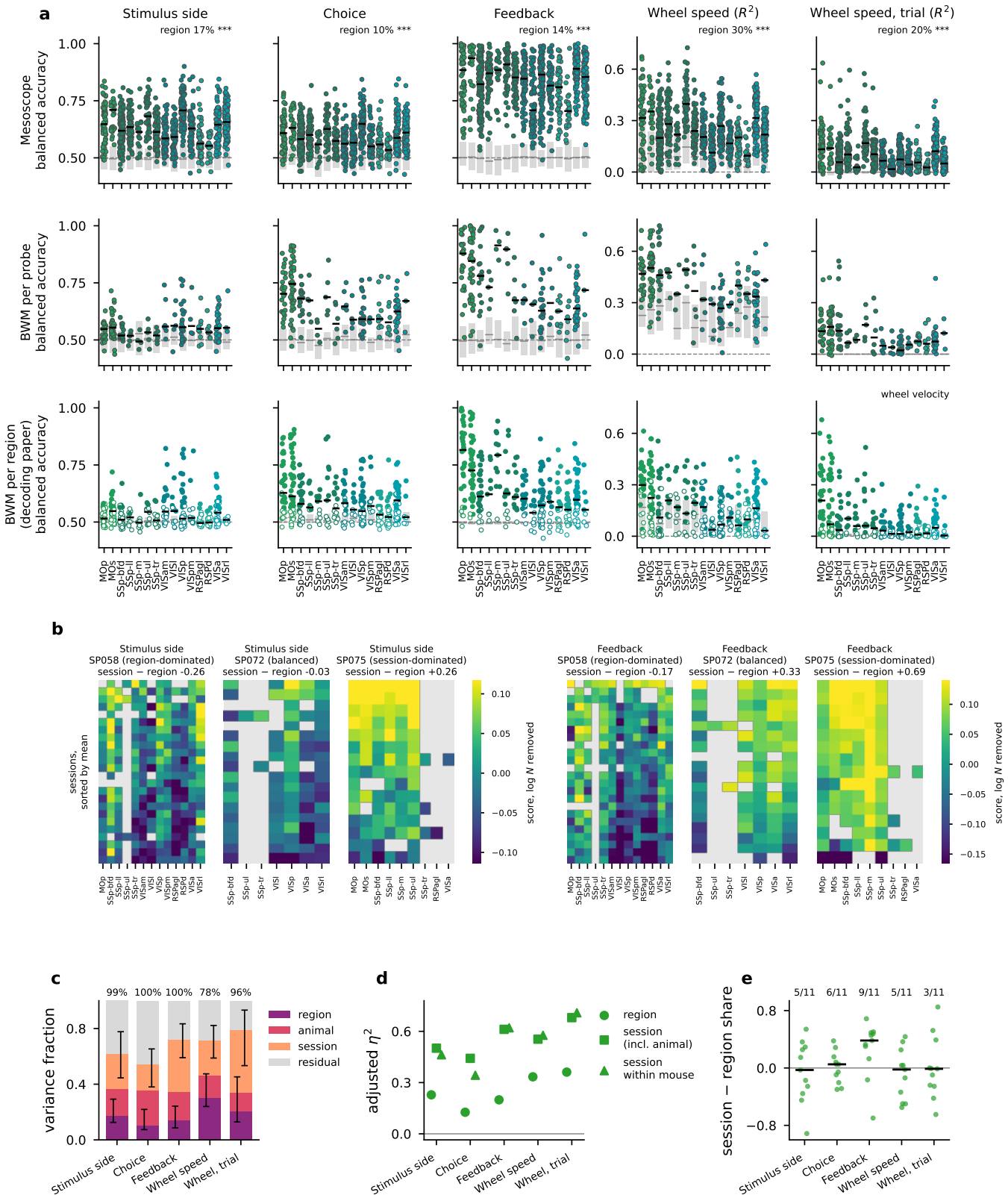



**Figure 1: Variance of decoding across regions and sessions (all neurons).** The 15 regions with ≥3 mesoscope sessions and BWM probes. Methods: *Region and session variance*. **a** Scores per region (columns ordered as in the Allen structure tree; Allen colours; bar: median; dashed: chance; grey band and line: 5–95th percentile and median of the baseline; for trial-specific wheel speed the shuffle baseline lies at $R^2 \approx 0$ and is hidden by the dashed line). Top: mesoscope, one dot per session and region from all its neurons there (149 sessions, 12 animals); baseline: shuffle control. Middle: BWM, one dot per probe from all its isocortex units, placed in the region where it enters the brain (198 probes, 93 animals; the majority region of its superficial units for 66% of probes); baseline: shuffle control. Bottom: BWM decoding paper, one dot per session and region from the region's units (346 pairs, 263 sessions, 116 animals); filled: $p < 0.05$ against pseudo-sessions (21–70% of dots); baseline: median of the pseudo-session null; wheel velocity in place of trial-specific wheel speed. Y axes shared within balanced accuracy and within $R^2$. Inset, top row: region share of variance from **c**. **b** Example mice (with ≥10 sessions and ≥7 regions): the most region-dominated, the most balanced and the most session-dominated by **e**, averaged over stimulus side and feedback. Each matrix: one row per session (sorted by its mean), one column per region (Allen order), colour: score with log $N$ regressed out (shared scale per variable), grey: region not imaged in that session. Horizontal stripes are session effects, vertical stripes region effects; columns that rank differently between mice are the region × mouse effect. **c** Mesoscope variance fractions from a mixed model, score ∼ log $N$ + (1|region) + (1|animal) + (1|session). Error bars: 95% intervals from resampling whole animals (200 refits), for the region share (left bar, at the top of the region block) and the session plus animal share (right bar, at the top of the session block); numbers above: fraction of resamples in which session plus animal exceeds region. Region: stimulus side 17%, choice 10%, feedback 14%, wheel speed 30%, wheel, trial 20%; session plus animal: stimulus side 45%, choice 44%, feedback 58%, wheel speed 41%, wheel, trial 59%. The region component is significant for every variable (likelihood-ratio test, $p < 0.001$). **d** The same question as **c**, do scores differ more between regions or between sessions, but without a statistical model, using only reshuffling: adjusted $\eta^2$ of region (circles; region labels shuffled within sessions), of session (squares; session labels shuffled within regions, across mice, so this includes differences between mice) and of session within mouse (triangles; shuffled within region and mouse, so differences between mice are removed, the model-free counterpart of **e**), 0 = chance. Filled: $p < 0.05$. **e** The same comparison within each mouse: mixed model score ∼ log $N$ + (1|region) + (1|session) fitted separately per mouse with ≥4 sessions (11 mice); dots: session share minus region share per mouse; bar: median; numbers above: mice with a larger session share.